PF LAB TASKS

Programming Fundamentals - Assignment

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Section: C

Roll No: 50

Submitted to: Dr Asad

Pf lab Task.

Task 1:-

(i) output
A
B

(ii) output
65
66

(iii) output
A
B

Task 2:-

Take two integers from user And Apply Arthmetic operation.

```
#include <iostream>
using namespace std;
int main() {
int a, b;
cout<<" enter the first number";
cout<<"
cin>>a;
cout<<" enter the second number";
cin>>B;
cout<<" Addition ="<<a+b<<endl;
cout<<" Substraction ="<<a-b<<endl;
cout<<" Multiplication =" a x b<<endl;
cout<<" Division =" a/b <<endl;
return 0;
}
```

Task 3 :-

output of we have enters 10, 20, 30

10
20
30
30
20
10

Task 4 :-

```
#include <iostream>
#include <cmath>
using namespace std;
int main(){
double x1, y1, x2, y2, d;
cout << "enter x1, y1 x2 y2";
cin >> x1 >> y1 >> x2 >> y2;
d = sqrt(pow(x2-x1, 2) + power(y2-y1, 2));
cout << " distance = " << d << endl;
 return 0;
}
```

Task 5:-

```
#include<iostream>
using namespace std;
int main() {
double r, length, width, base, height;
const double PI = 3.14159;
// 1. Area of circle
cout<<"enter radius of circle:";
cin>>r;
double area circle = PI*r*r;
cout<<" Area of circle = "<<area circle<<endl;
// 2. Area of Rectangle
cout<<" Enter length and width of Rectangle:";
cin>>length>>width;
double area Rect = length*width;
cout<<"Area of Rectangle = "<<area Rect<<end;
// 3. Area of triangle
cout<<"enter base and height of triangle:";
cin>>base>>height;
double area tri = base*height*0.5;
cout<<" area of triangle = "area tri<<endl;
return 0;
}
```

Task 6:-

```
#include<iostream>
using namespace std;
int main() {
double price1 = 12.95;
double price2 = 24.95;
double price3 = 6.95;
double price4 = 14.95;
double price5 = 3.95;
double subtotal = price1 + price2 + price3 + price4 + price5;
double salestax = subtotal * 0.06; // 6% tax;
double total = subtotal + salestax;
cout
cout<<"price of item1 = $"<<price1<<endl;
cout<<"price of item2 = $"<<price2<<endl;
cout<<"price of item3 = $"<<price3<<endl;
cout<<"price of item4 = $"<<price4<<endl;
cout<<"price of item5 = $"<<price5<<endl;
cout<<"----------"<<endl;
cout<<"subtotal = $"<<subtotal<<endl;
cout<<"sales tax (6%) = $"<<salestax<<endl;
cout<<"total = $"<<total<<endl;
return 0;
}
```

Bonus task:-

```
#include<iostream>
using namespace std;
int main() {
int num = 42339 // you can change value here e.g 12345
cout<<"Number is:"<<num<<endl;
int d1 = num/10000;
int d2 = (num/1000)%10;
int d3 = (num/100)%10;
int d4 = (num/10)%10;
int d5 = num%10;
cout<<d1<<" "<<d2<<" "<<d3<<" "<<d4<<" "<<d5<<endl;
return 0;
}
```

Home task 2:-

```
#include<iostream>
using namespace std;
int main() {
char ch;
cout<<"enter an alphabet in small caps: ";
cin>>ch;
// conver using ASCII
char upper = ch-32;
cout<<"upercase:"<<upper;
return 0;
}
```

Home task 1:-

```
#include<iostream>
using namespace std;
int main() {
int var1 = 10;
int var2 = 20;
cout<<"Before swap:"<<endl;
cout<<" var1 = "<<var1<<endl;
cout<<"var2 = "<<var2<<endl;
// swapping logic
int tem = var1;
 var1 = var2;
 var2 = temp;
 cout <<"\n After swap:" <<end;
 cout <<"var1 = "<<var1<<"(value of var2)"endl;
 cout<<"var2 = "<<var2<<"(value of var1)" endl;
 return 0;
}
```

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new*

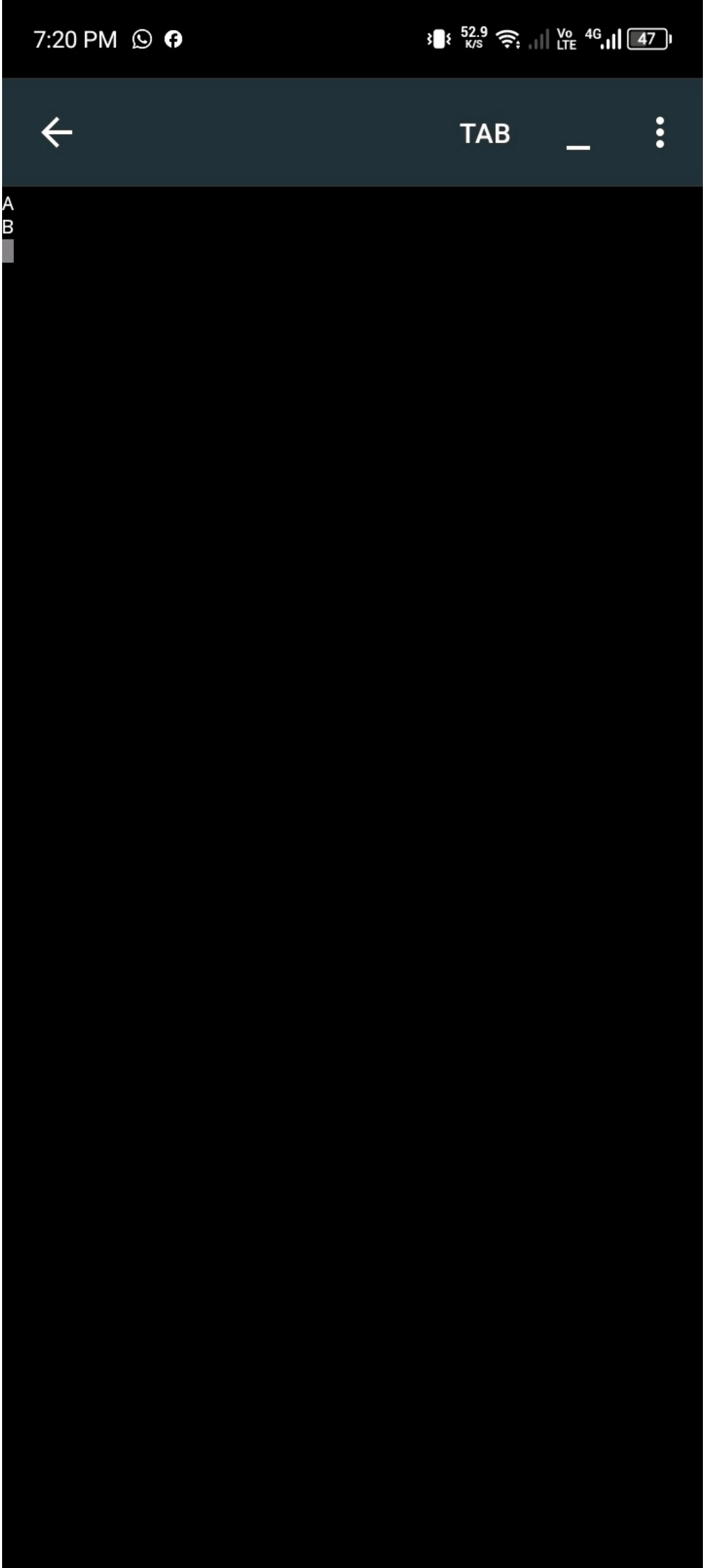
new* new new

```
1  #include <iostream>
2  using namespace std;
3  int main( ){
4     char letter;
5     letter='A'
6     cout<<letter<<'/n';
7     letter='B'
8     cout<<letter<<'/n';
9     cin>>letter;
10    return 0;
11 }
```

: ; " ' #

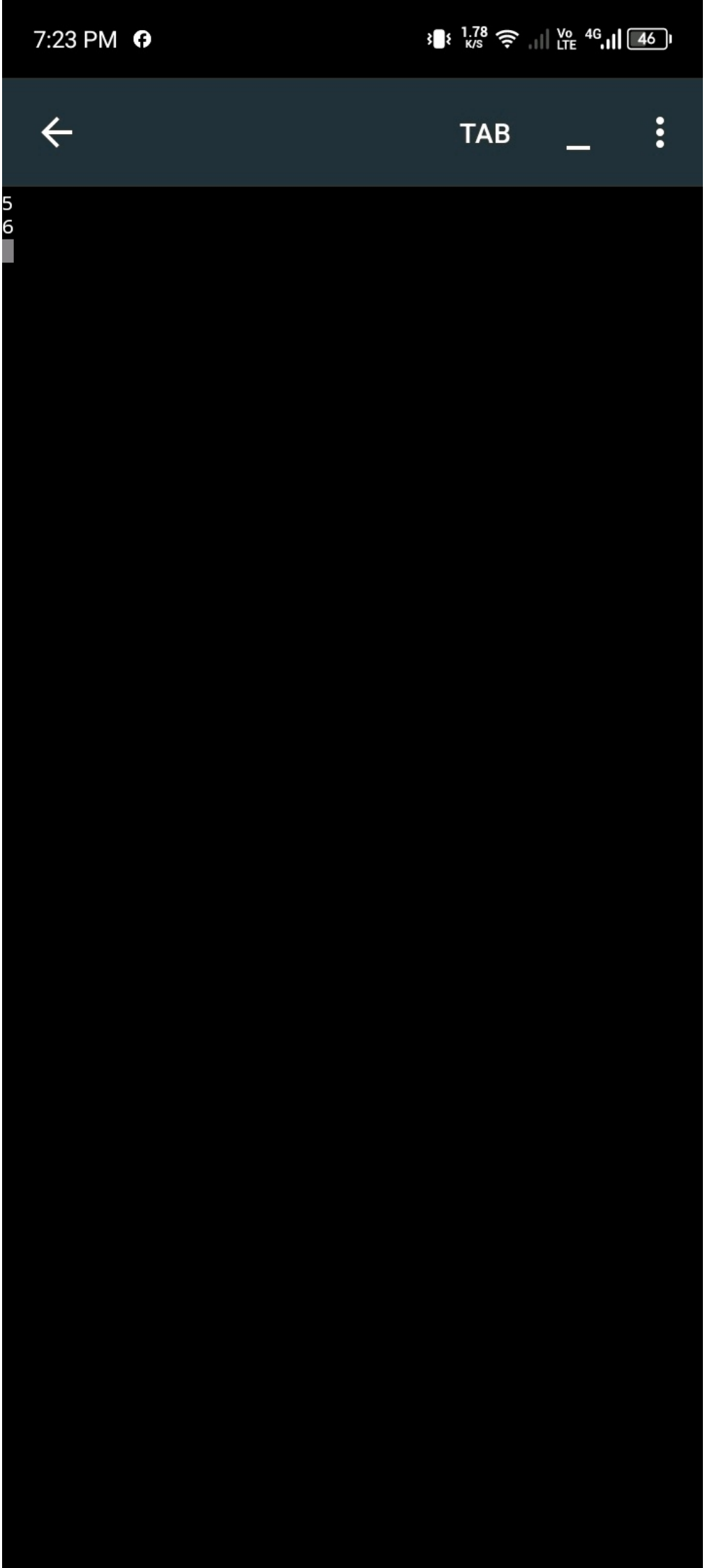
new* new new

```
1  #include <iostream>
2  using namespace std;
3  int main( ){
4     char letter;
5     letter='A';
6     cout<<letter<<"\n";
7     letter='B';
8     cout<<letter<<"\n";
9     cin>>letter;
10    return 0;
11 }
```


7:20 PM
52.9 K/S
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47
TAB
A
B

(invalid)* new new

```
1  #include <iostream>
2  using namespace std;
3  int main( ){
4     char letter;
5     letter='65';
6     cout<<letter<<"\n";
7     letter='66';
8     cout<<letter<<"\n";
9     cin>>letter;
10    return 0;
11 }
```


7:23 PM
1.78 K/S
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46
TAB
5
6

7:36 PM
196 B/S
Vo LTE 4G 44
(invalid)*
SAF:Downloads/(...
(invalid)* new new

```cpp
#include <iostream>
using namespace std;

int main(){
   int a, b;
   cout << "Enter first number: ";
   cin >> a;
   cout << "Enter second number: ";
   cin >> b;

   cout << "Addition = " << a + b << endl;
   cout << "Subtraction = " << a - b << endl;
   cout << "Multiplication = " << a * b <<
endl;
   cout << "Division = " << a / b << endl;

   return 0;
}
```

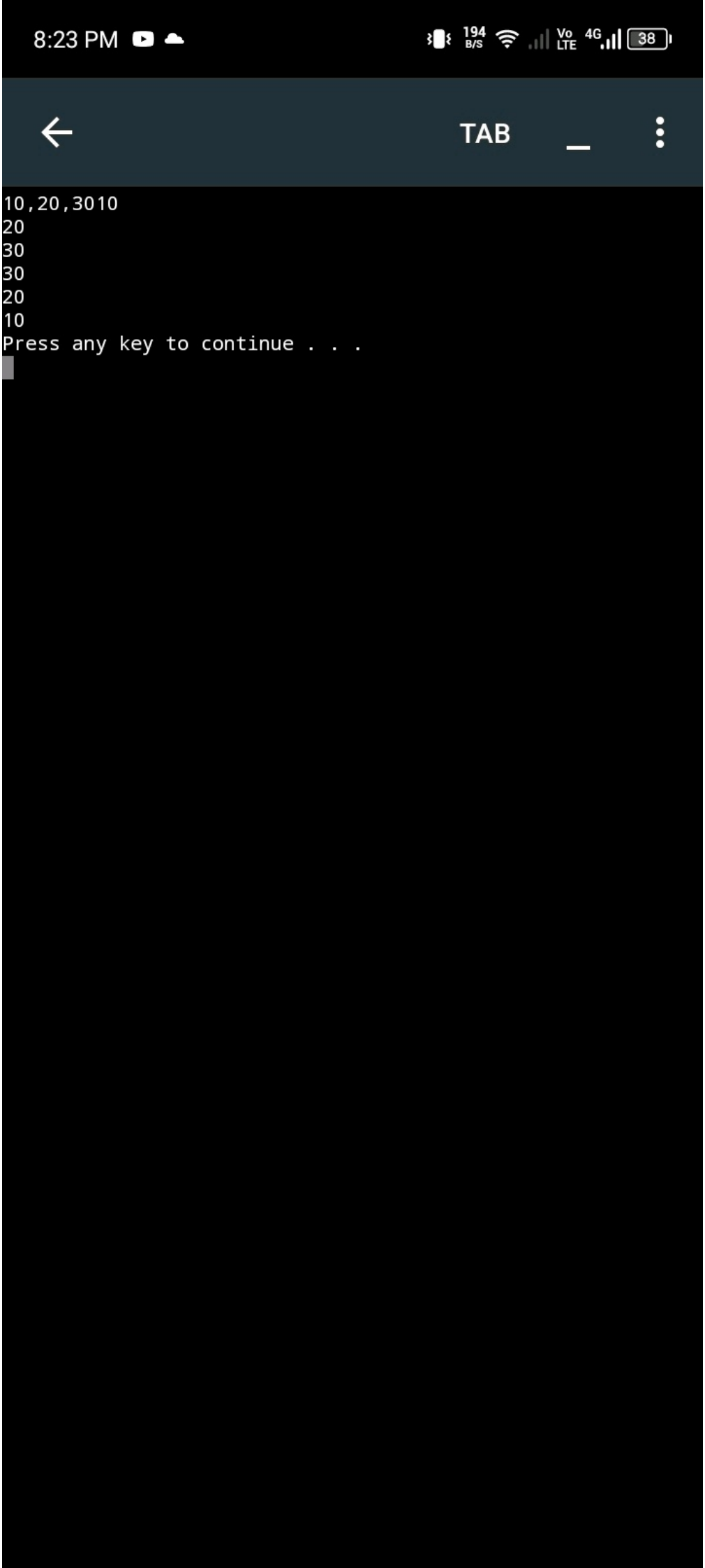
Tab | { | } | : | ; | "

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(invalid)* new* new*

```
1  #include<iostream>
2  using namespace std;;
3  int main( ){
4     int val1=10;
5     int val2=20;
6     int val3=30;
7     cout<<"10,20,30";
8
   cout<<val1<<"\n"<<val2<<"\n"<<val3<<"\n";
9
   cout<<val3<<"\n"<<val2<<"\n"<<val1<<"\n";
10    system("pause");
11    return 0;
12 }
13
```

) [ ] = - +


8:23 PM
194 B/S
Vo LTE
4G
38
TAB
10,20,3010
20
30
30
20
10
Press any key to continue . . .

11:23 PM

(invalid)* new* new* new

```cpp
#include <iostream>
using namespace std;

int main(){
    double r, length, width, base, height;
    const double PI = 3.14159;

    // 1. Area of Circle
    cout << "Enter radius of Circle: ";
    cin >> r;
    double areaCircle = PI * r * r;
    cout << "Area of Circle = " << areaCircle << endl << endl;

    // 2. Area of Rectangle
    cout << "Enter length and width of Rectangle: ";
    cin >> length >> width;
    double areaRect = length * width;
    cout << "Area of Rectangle = " << areaRect << endl << endl;

    //Area of Triangle
    cout << "Enter base and height of Triangle: ";
    cin >> base >> height;
    double areaTri = base * height * 0.5;
    cout << "Area of Triangle = " << areaTri << endl;

    return 0;
}
```

= | - | + | * | / | ^

11:31 PM 4.84 K/S Vo LTE 4G 51


```cpp
#include <iostream>
using namespace std;

int main(){
   double price1 = 12.95;
   double price2 = 24.95;
   double price3 = 6.95;
   double price4 = 14.95;
   double price5 = 3.95;

   double subtotal = price1 + price2 + price3 + price4 + price5;
   double salesTax = subtotal * 0.06; // 6% tax;
   double total = subtotal + salesTax;

   cout << "Price of item 1 = $" << price1 << endl;
   cout << "Price of item 2 = $" << price2 << endl;
   cout << "Price of item 3 = $" << price3 << endl;
   cout << "Price of item 4 = $" << price4 << endl;
   cout << "Price of item 5 = $" << price5 << endl;
   cout << "----------------------" << endl;
   cout << "Subtotal = $" << subtotal << endl;
   cout << "Sales Tax (6%) = $" << salesTax << endl;
   cout << "Total = $" << total << endl;

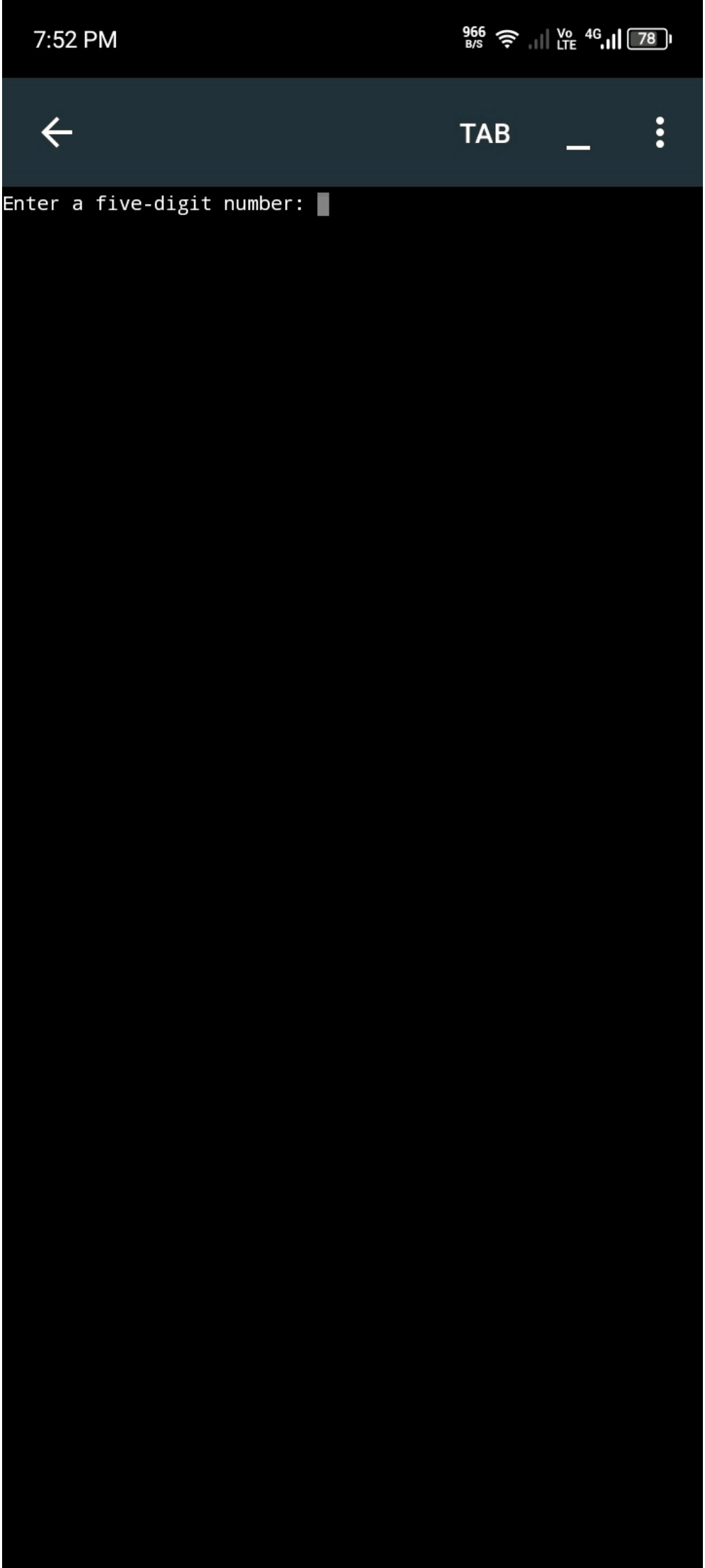
   return 0;
```

```cpp
#include <iostream>
using namespace std;

int main() {
    int num;
    cout << "Enter a five-digit number: ";
    cin >> num;

    int d1 = num / 10000;
    int d2 = (num / 1000) % 10;
    int d3 = (num / 100) % 10;
    int d4 = (num / 10) % 10;
    int d5 = num % 10;

    cout << d1 << "   " << d2 << "   " << d3 << "
   " << d4 << "   " << d5;
    return 0;
}
```


7:52 PM
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TAB
Enter a five-digit number:

```cpp
#include <iostream>
using namespace std;

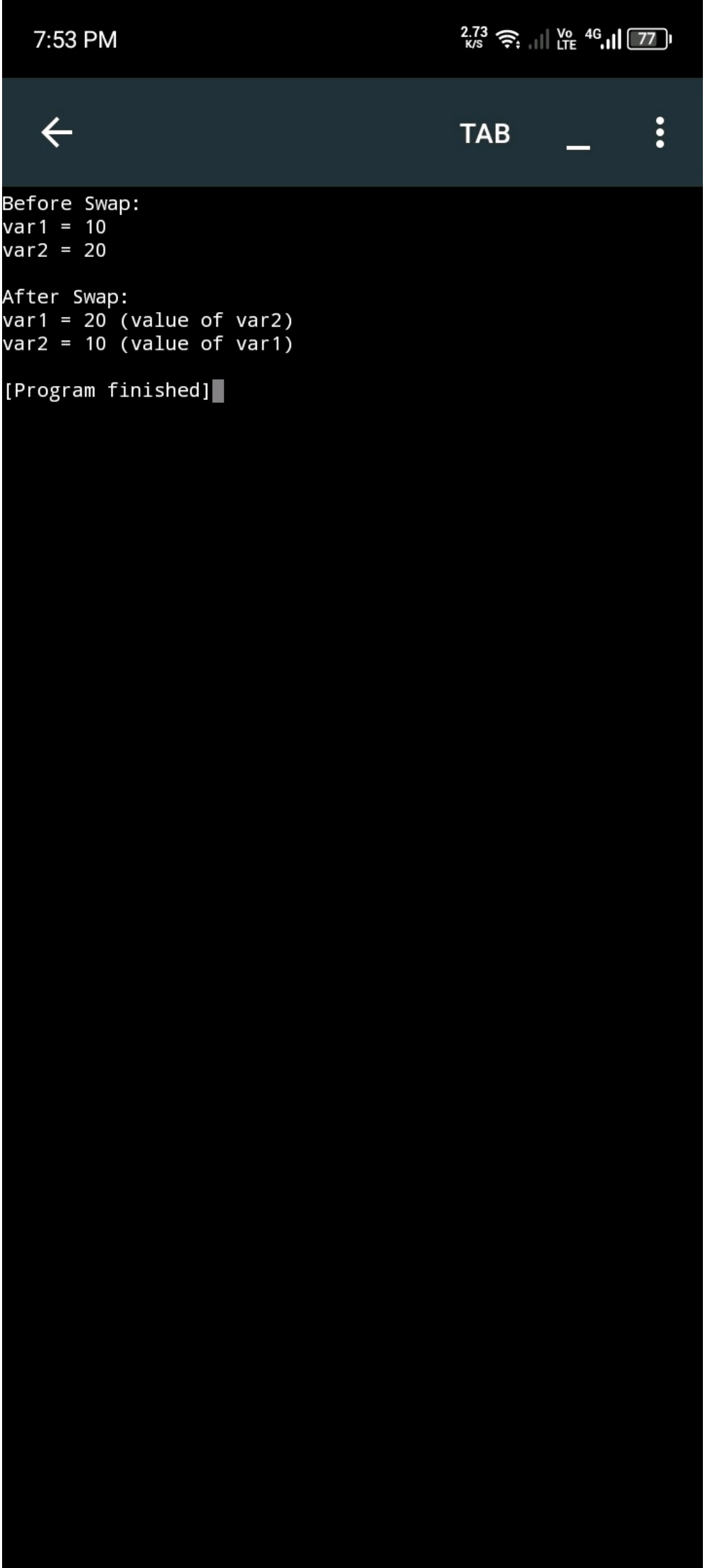
int main() {
    int var1 = 10;
    int var2 = 20;

    cout << "Before Swap:" << endl;
    cout << "var1 = " << var1 << endl;
    cout << "var2 = " << var2 << endl;

    // Swapping Logic
    int temp = var1;
    var1 = var2;
    var2 = temp;

    cout << "\nAfter Swap:" << endl;
    cout << "var1 = " << var1 << " (value of var2)" << endl;
    cout << "var2 = " << var2 << " (value of var1)" << endl;

    return 0;
}
```


7:53 PM
2.73 K/S
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77
TAB
Before Swap:
var1 = 10
var2 = 20
After Swap:
var1 = 20 (value of var2)
var2 = 10 (value of var1)
[Program finished]

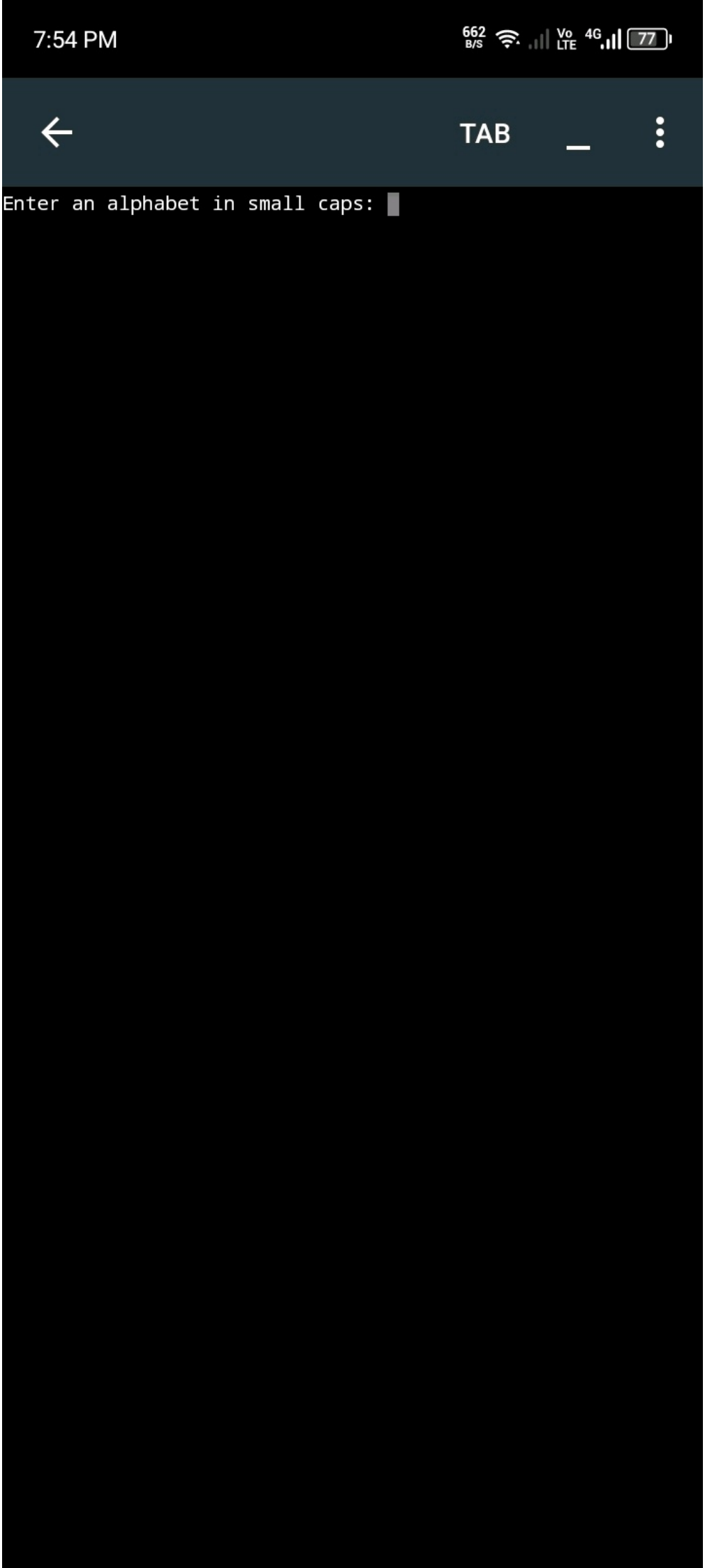
```cpp
#include<iostream>
using namespace std;

int main() {
    char ch;
    cout << "Enter an alphabet in small caps: ";
    cin >> ch;

    // Convert using ASCII
    char upper = ch - 32;

    cout << "Uppercase: " << upper;

    return 0;
}
```


7:54 PM
662 B/S
Vo LTE
4G
77
TAB
Enter an alphabet in small caps: